FAUCON

CIF 2026 · BUSINESS HACKATHON · CAMEROON



FAUCON

The control tower for African fleets.

Faucon is a Cameroonian platform that connects vehicle GPS trackers to a web and mobile application, giving companies real-time tracking, alerts and trip history.

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Douala, Cameroon
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01 · THE PROBLEM

Without visibility, every vehicle is a blind spot.

In Cameroon and Central Africa, entire fleets operate without reliable data: delivery vehicles, motorcycles, trucks, construction equipment and company cars. Operators make decisions with phone calls, spreadsheets and messages.

Blind spots

It is impossible to know where a vehicle is, whether it is on route, or whether it is immobilized.

Late alerts

A speeding event or a geofence exit reported too late no longer protects anyone.

Decisions in the dark

The data exists but remains scattered. It is impossible to compare, prove or anticipate.

> **“The real problem is not knowing where the vehicle is. It is knowing whether it is safe, on route, and who must act now.”**

Imported solutions are often expensive, dependent on perfect connectivity and designed for European roads. The Cameroonian context demands another approach: resilience, controlled cost, remote diagnostics and local support.

02 · THE PROBLEM IN NUMBERS

A measurable problem, not a hunch.

Cameroon drives, delivers, builds and moves. But the information that could protect these vehicles and optimize these journeys remains scarce, expensive or late.

2,870

estimated road deaths

Cameroon, 2021. That is 10.6 deaths per 100,000 people according to the WHO.

Source: WHO, Global status report on road safety 2023.

675,000

vehicles on the road

Estimated fleet in 2014, including around 43% motorcycles. The fleet has grown sharply since then.

Source: UNECE / CIT, Cameroon vehicle inspection, 2019.

41.9%

of the population online

12.6 million internet users out of ~29 million people: 58% remain offline.

Source: DataReportal, Digital 2026: Cameroon.

63 %

higher transport costs

In Africa, transport costs are 63% higher than in developed countries.

Source: African Development Bank, 2013.

WHAT THIS MEANS ON THE GROUND

A logistics manager in Douala runs fifteen trucks with WhatsApp and a spreadsheet. He learns that a vehicle left its zone the next day, when the customer calls. He cannot prove the route, anticipate downtime, or compare drivers with reliable data.

THE CAMEROONIAN CONSTRAINT

- **Unreliable connectivity**
  The tracker must keep collecting when the network is down.
- **Cost of access**
  Pricing must remain compatible with local SMEs.
- **Demanding terrain**
  Heat, dust, vibration and unstable power.

03 · THE SOLUTION

One platform. One device. The whole fleet.

Faucon brings together the tracker, the network, the server and the web/mobile interfaces. One chain, from the field to the decision.

Real-time position

Smooth map, real heading, speed, battery and device signal.

Field alerts

Speeding, geofence exit, impact, tracker removal.

History & replay

Trips, stops, distances and evidence available day by day.

Diagnostics

GPS, network, battery and device health, remotely.

Web + mobile

The same operational truth at the desk and in the field.

Owner API

Customer teams integrate the data into their own tools.

WHAT SETS US APART

- ✓ **Built for the field**
  Adaptive modes, local journal, recovery after network loss.
- ✓ **Controlled cost**
  Local device, web platform included, no map API key.
- ✓ **Data sovereignty**
  Infrastructure owned, hosted and operated by Faucon.

ALREADY OPERATIONAL

A deployed platform, an Android app, a documented API and a real tracker reporting data.

Tracker
ESP32-C3 · GPS · mobile network

Transport
MQTT + WebSocket

Web
Next.js · OSM map

Mobile
Flutter · Android

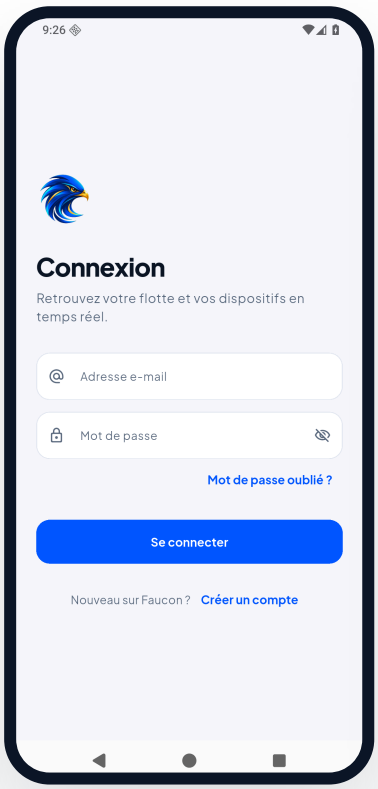
04 · THE MOBILE EXPERIENCE

A simple, blue, human application.

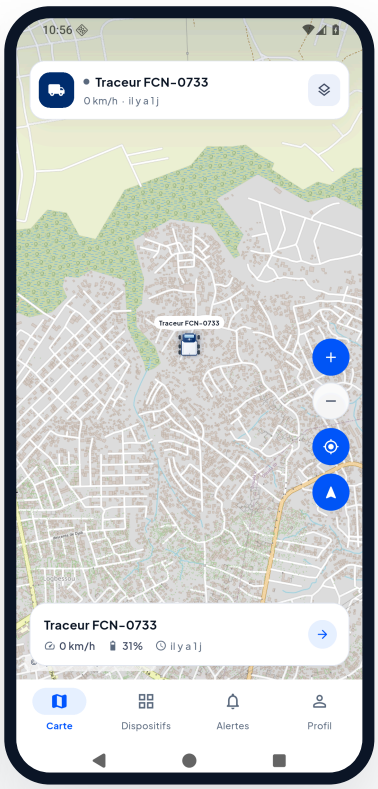
Clear onboarding, quick login, readable map: the app was designed for field use, even by people unfamiliar with GPS tools.



First launch



Login



Live map

TRACKING

Truck oriented to its heading, smoothed position, preserved zoom.

ALERTS

Badge, filters, acknowledgement and context for each event.

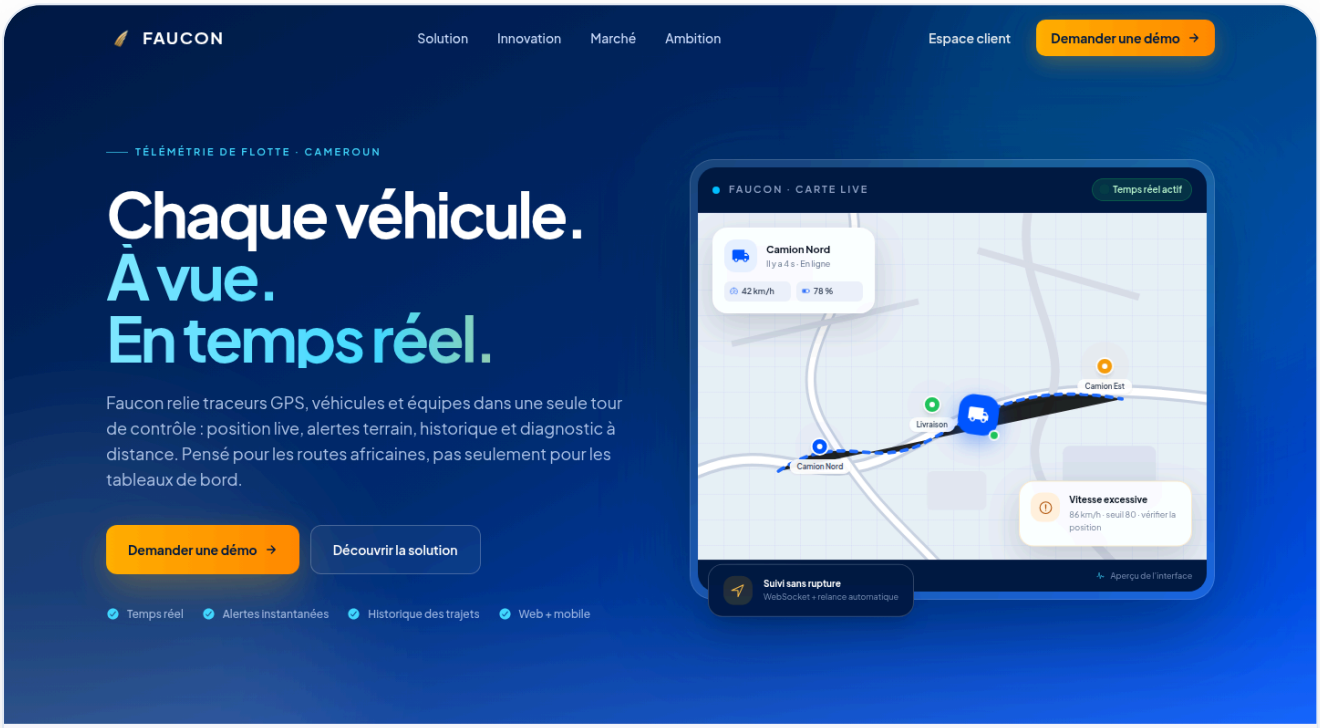
INTEGRATION

API keys to connect data to your own tools.

05 · THE WEB PLATFORM

The control tower, on the big screen.

The web dashboard provides a consolidated view: fleet, alerts, trackers, users and the live map. The public landing page presents the product and documents the owner API for customer technical teams.



OWNER API

Only the useful data, read-only

Raw positions, history, alerts and device status. One key per integration, revocable, limited to the customer's vehicles.

SAFETY

Designed for integrations

Scoped API key, SHA-256 hash, no internal data exposed, multi-language documentation.

06 · HOW IT WORKS

From device to decision-maker, a controlled chain.

The tracker collects, the network transports, Faucon processes and broadcasts. Every link is designed to keep producing useful information when the field does not cooperate.

Faucon tracker
GPS · heading · battery
mobile network

→

MQTT
lightweight
resilient transport

→

Faucon API
positions · alerts
diagnostics

→

Web + mobile
map · history
customer integrations

Local journal
positions stored
offline

↺

Diagnostics
device health
every 5 minutes

→

Alerts
thresholds · zones
impacts

→

Decision
protect · prove
optimize

Adaptive modes

WORK, MOVE, STANDBY: reporting frequency adapts to movement and energy.

Network resilience

Positions are stored locally and republished when the network returns.

Fused heading

GPS + gyroscope for a stable direction, even at low speed.

Embedded diagnostics

The device reports its real state: battery, GPS, network, memory and reboots.

07 · THE OPPORTUNITY

The digitalization of African fleets is only beginning.

Urbanization, logistics growth and the need for proof create structural demand: knowing where vehicles are, how they are used, and what deserves intervention.

SEGMENT 1

Transport & logistics

Trucks, vans, urban and intercity distribution.

SEGMENT 2

Enterprises & construction

Sites, machinery, company vehicles and mobile assets.

SEGMENT 3

Insurance & risk

Timestamped history, proof of use, prevention and claims handling.

SEGMENT 4

Public sector & NGOs

Health, water, agriculture, humanitarian: fleets that matter where connectivity is imperfect.

A REAL ADDRESSABLE MARKET

The Cameroonian fleet exceeds **675,000 vehicles** (2014), with a significant share of professional fleets. A conservative 10% fleet assumption already represents **67,500 vehicles**.

REVENUE SCENARIO

At XAF 7,500 / vehicle / month, this segment would represent a potential of **XAF 6.1 billion per year**. This is an internal order of magnitude, not a forecast.

GO-TO-MARKET

Field pilots on real fleets, with value measurement.

Local partners : carriers, insurers, integrators and schools.

Network effect : every equipped fleet becomes a reference.

08 · BUSINESS MODEL

A subscription that pays for itself in the field.

Faucon does not just sell a device: it sells reduced risk, less fuel diversion and less lost time. The model is easy for a Cameroonian SME to understand.

WHAT THE CUSTOMER PAYS FOR

1. **The tracker**
   Installed and configured device, with SIM and power supply.
2. **The subscription**
   Web + mobile platform, alerts, history, support and updates.
3. **API integration**
   Optional for companies that want the data in their ERP.

EXAMPLE SCENARIO

A distribution company in Douala equips **15 trucks**. Its first goal is to reduce unauthorized trips and downtime.

CALCULATION ASSUMPTIONS (ILLUSTRATIVE)

Fleet	**15 trucks**
Monthly distance	**2,500 km / truck**
Consumption	**22 L / 100 km**
Diesel price	**XAF 840 / L**
Target saving	**5%**
Subscription	**XAF 7,500 / truck / month**

Monthly fleet fuel	**XAF 6,930,000**
Monthly saving (5%)	**XAF 346,500**
Monthly subscription	**XAF 112,500**
Net monthly gain	**XAF 234,000**
Net annual gain	**XAF 2,808,000**
Hardware payback	**~4.8 months**

Faucon working assumptions, to be validated with pilot customers. Actual gains depend on usage, fuel and avoided theft/diversion.

09 · IMPACT & AMBITION

Become the reference in telematics in Central Africa.

EXPECTED IMPACT

- **Safety**
  Fewer lost vehicles, fewer risky behaviors, faster response.
- **Productivity**
  Decisions based on data, not phone calls.
- **Trust**
  Usable history for customers, insurers and partners.
- **Local skills**
  A product designed, developed and operated in Cameroon.

ROADMAP

2026	**Consolidate** Make the tracker reliable, industrialize the device, deploy the first pilots.
2027	**Scale up** Partner network, support, new verticals.
NEXT	**Central Africa** Adapt reporting, languages and support to each market's reality.

> "We do not sell a map. We sell the ability to know, prove and act."

10 · THE TEAM

Two skills, one school, a complete product.

Faucon is led by two students from the École Nationale Supérieure Polytechnique de Douala, with complementary software and hardware skills.

TC

TSEFACK Calvin Klein

SOFTWARE ENGINEERING · ENSPD

API design, web and mobile applications, tracker-to-app communication, deployment on a Contabo VPS.

École Nationale Supérieure Polytechnique de Douala
Year 5 · Software Engineering

DF

DOMCHE Freedy Jores

NETWORK & TELECOMMUNICATIONS ENGINEERING · ENSPD

Electronics, device design, IoT integration, sensors, power and tracker connectivity.

École Nationale Supérieure Polytechnique de Douala
Network & Telecommunications Engineering

SOFTWARE

API, web, mobile, real time, security, deployment.

HARDWARE

Device, board, sensors, power, network.

FIELD

Understanding of local uses, costs and constraints.

11 · CONCLUSION

The field has its constraints. Faucon has its answer.

A tracker built to last, a web and mobile platform, an API for partners, a local team: Faucon turns raw GPS data into useful decisions for African fleets.

WHAT WE ALREADY HAVE

- ✓ **Deployed platform**
  API and site in production on a VPS.
- ✓ **Android application**
  Onboarding, map, alerts, tracker details, API keys.
- ✓ **Real tracker**
  Communication between device and application validated.
- ✓ **Documented API**
  Secure read-only customer integration.

WHAT WE ARE LOOKING FOR

Field pilots, distribution partners, industrial mentors and support to industrialize the device and scale.

Let's build the next step.

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Douala · Cameroon

Sources: WHO, *Global status report on road safety 2023 — Cameroon* (2,870 estimated deaths, 2021) · DataReportal, *Digital 2026: Cameroon* (29 M connexions mobiles, 12.6 M internautes, 41.9%) · UNECE / CIT, *Vehicle inspection and approval in Cameroon*, 2019 (675,000 vehicles in 2014, 43% motorcycles) · African Development Bank, 2013 (transport costs in Africa +63%) · Market and ROI assumptions: internal Faucon work, to be validated in pilots.